

LAB 11

Flutter: API integration

API: REST HTTP API (OpenRouter AI)

Openrouter_service.dart

API imports & API Key:

```
1  import 'dart:convert';
2  import 'package:http/http.dart' as http;
3
4  // Replace with your OpenRouter API key
5  const _apiKey = 'YOUR_OPENROUTER_API_KEY';
6  const _baseUrl = 'https://openrouter.ai/api/v1/chat/completions';
7  const _model = 'mistralai/mistral-7b-instruct';
8
```

API Key:

```
4  // OpenRouter API key
5  const _apiKey = 'YOUR_OPENROUTER_API_KEY';
6  const _baseUrl = 'https://openrouter.ai/api/v1/chat/completions';
7  const _model = 'mistralai/mistral-7b-instruct';
```

Chat with bot using API

```
11 Future<String> chatWithAI(List<Map<String, String>> history) async {
12   try {
13     final messages = [
14       {'role': 'system', 'content': _systemPrompt},
15       ...history,
16     ];
17
18     final response = await http.post(
19       Uri.parse(_baseUrl),
20       headers: {
21         'Authorization': 'Bearer $_apiKey',
22         'Content-Type': 'application/json',
23         'HTTP-Referer': 'https://persist-app.com',
24         'X-Title': 'Persist',
25       },
26       body: jsonEncode({
27         'model': _model,
28         'messages': messages,
29         'max_tokens': 500,
30         'temperature': 0.7,
31       })),
32     );
33
34     if (response.statusCode == 200) {
35       final data = jsonDecode(response.body);
36       return data['choices'][0]['message']['content'] as String;
37     }
38     return "I'm having trouble connecting right now. Please try again in a mo
39   } catch (_) {
40     return "I'm having trouble connecting right now. Please try again in a mo
41   }
```

```
44 Future<List<Map<String, dynamic>>> generateGoalPlan(
45   String goalName, int days, String category) async {
46   try {
47     final prompt = '''Create a structured $days-day plan for the goal: "$goal
48
49 Return ONLY a JSON array with this exact format, no markdown:
50 [
51   {"dayNum": 1, "title": "Day 1 – Introduction", "tasks": ["Task 1", "Task 2"
52   {"dayNum": 2, "title": "Day 2 – Building Foundation", "tasks": ["Task 1", "
53 ]
54
55 Rules:
56 - Generate exactly $days days
57 - Each day has exactly 3 tasks
58 - Tasks should be specific and actionable
59 - Titles should be progressive and motivating
60 - Return pure JSON array only, no other text''';
61
62   final response = await http.post(
63     Uri.parse(_baseUrl),
64     headers: {
65       'Authorization': 'Bearer $_apiKey',
66       'Content-Type': 'application/json',
67       'HTTP-Referer': 'https://persist-app.com',
68       'X-Title': 'Persist',
69     },
70     body: jsonEncode({
71       'model': _model,
72       'messages': [
73         {'role': 'user', 'content': prompt}
74       ],
75       'max_tokens': 3000,
76       'temperature': 0.4,
77     })),
78   );
79
80   if (response.statusCode == 200) {
81     final data = jsonDecode(response.body);
82     var content = data['choices'][0]['message']['content'] as String;
```

```

83     content = content
84         .replaceAll("`json`, ``")
85         .replaceAll("`", ``)
86         .trim();
87
88     final List<dynamic> parsed = jsonDecode(content);
89     return parsed.cast<Map<String, dynamic>>();
90 }
91 } catch (_) {}
92 return null;
93 }

```



